Program 1

```
#include <iostream>
using namespace std ;

class First {
    protected:
        int a ;
    public:
        First(int x = 5)
            { a = x ; }
        int getVal()
            { return a ; } //a has value of 5, will return 5
} ;

class Second : public First {
    private:
        int b ;
    public:
        Second(int y = 4)
            { b = y ; }
        int getVal()
            { return b ; } //b has value of 4, will return 4
} ;

int main() {
    First object1 ;
    Second object2 ;

    cout << object1.getVal() << endl ;
    cout << object2.getVal() << endl ;

    return 0 ;
}
```

This program will output 5 & 4 on separate lines.

Program 2

```
#include <iostream>
using namespace std ;

class First {
    protected:
        int a ;
    public:
        First(int x = 5)
            { a = x ; }
        void twist()
            { a *= 3 ; }
        int getVal()
            { twist() ; return a ; }
} ;

class Second : public First {
    private:
        int b ;
    public:
        Second(int y = 3)
            { b = y ; }
        void twist()
            { b *= 10 ; }
} ;

int main() {
    First object1 ;
    Second object2 ;

    cout << object1.getVal() << endl ;
    cout << object2.getVal() << endl ;

    return 0 ;
}
```

// when the getVal() function is called it twists a, and returns a. a will equal 15.

// The getVal() is inherited from First and will be called resulting in twist a return a.

// outputs 15
// outputs 15

This program will output 15 & 15 on separate lines.

Program 3

```
#include <iostream>
using namespace std ;

class First {
    protected:
        int a ;
    public:
        First(int x = 5)
            { a = x ; }
        virtual void twist()
            { a *= 3 ; }
        int getVal()
            { twist() return a ; }
} ;

class Second : public First {
    private:
        int b ;
    public:
        Second(int y = 4)
            { b = y ; }
        virtual void twist()
            { b *= 6 ; }
} ;

int main() {
    First object1 ;
    Second object2 ;

    cout << object1.getVal() << endl ;
    cout << object2.getVal() << endl ;

    return 0 ;
}
```

// twist a, then return. a=15

// twist is redificed and when getVal() is called then it twists b but still returns a untwisted.

// outputs 15

// outputs 5

This program will output 15 & 5 on separate lines.

Program 4

```
#include <iostream>
using namespace std ;

class First {
    protected:
        int a ;
    public:
        First(int x = 3)
            { a = x ; }
        virtual void twist()
            { a *= 4 ; } // a = 12
        int getVal() {
            twist() ;
            cout << "Inside getVal(), a: " << a << endl ;
            return a ;
        }
} ;

class Second : public First {
    private:
        int b ;
    public:
        Second(int y = 8)
            { b = y ; }
        virtual void twist() {
            cout << "Inside twist(Second - before, b): " << b << endl ;
            b *= 10 ;
            cout << "Inside twist(Second - after, b): " << b << endl ;
        }
} ;

int main() {
    First object1 ;
    Second object2 ;

    cout << object1.getVal() << endl ;
    cout << object2.getVal() << endl ;

    return 0 ;
}
```

// again this function redefines twist() but still does not return b.

Output:

Inside getVal(), a: 12
12
Inside twist (Second - before, b): 8
Inside twist (Second - after, b): 80
Inside getVal(), a: 3
3

Program 5

```
#include <iostream>
using namespace std ;

class Base {
    protected:
        int baseVar ;
    public:
        Base(int val = 6)
            { baseVar = val ; } // baseVar = 6
        int getVar()
            { return baseVar ; }
} ;

class Derived : public Base {
    private:
        int derivedVar ;
    public:
        Derived(int val = 11)
            { derivedVar = val ; } // derivedVar = 11
        int getVar()
            { return derivedVar ; }
} ;

int main() {

    Base *optr ;
    Derived object ;

    optr = &object ;
    cout << optr -> getVar() << endl ;

    return 0 ;
}
```

// optr now points to -> object
// will return getVar() for from Base class

The output for this project is: 6